

Towfiqul Islam

Senior Software Engineer

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SKILLS

Java (Spring Boot), PHP (Laravel), JavaScript (jQuery), MySQL, PostgreSQL, Oracle, Docker, REST API, AWS, AZURE, CLI, Tomcat, Postman, JUnit, Rest Assured, Mockito, Ubuntu, NGINX, OAuth2, KeyCloak, ActiveMQ, Kafka, Kubernetes

WORK EXPERIENCE

Senior Software Engineer (Full-time, Hybrid)

Exabyting 🔗

05/2024 – present
Dhaka, Bangladesh

- Worked with **bKash Limited** as an Augmented Engineer with their core Research and Engineering Team
- Designed scalable REST APIs using Spring Boot
- Optimized database queries + Redis caching to improve response time
- Architecture design & microservices implementation for the core transaction platform
- Implemented unit & integration testing
- Mentored junior engineers & led technical discussions
- Established CI/CD pipelines using GitHub Actions, Bash Script, and Docker
- Production monitoring using Grafana & Kibana
- Applied coding standards and principles with Domain Driven Design

Software Engineer (Full-time, Remote)

SELISE 🔗

11/2021 – 04/2024
Zürich, Switzerland

- Responsible for defining & developing software for tasks associated.
- Develop enhancements to existing products.
- Collaborate in a team environment to share best practices.
- Implement process improvements.
- Ensure software quality.
- Led a three-member group and delivered tasks within the deadline.
- Worked with a foreign client and established good communication, and discussed various technical and business topics.

Software Engineer (Full-time, Remote)

Exabyting 🔗

12/2019 – 11/2021
Dhaka, Bangladesh

PROJECTS

bKash (Customer App) 🔗

10/2025 – Present

Worked for **bKash Limited** 🔗 with their **core engineering team**. Engineered scalable high-performance **microservices** using **Java 11-21** and **Spring Boot**, leveraging modern concurrency patterns like **Virtual Threads** and **reactive programming** to build **resilient, fault-tolerant systems**. Managed robust cloud deployments via **AWS** while optimizing data access layers, including significant migrations from **Spring Boot 2 to 3** and **AWS SDK v1 to v2**. Drove continuous improvement by implementing **clean architecture** principles, developing internal automation tools, and performing rigorous performance tuning to support **high-traffic** applications within collaborative, **Agile** environments.

Tesenso Middleware

06/2022 – 12/2023

An IoT Based Project based on ThingsBoard

Engineered a robust, **multi-tenant IoT** platform for **Tesenso**  to manage **large-scale electrical vehicle** and **aerospace** data by extending **ThingsBoard** with custom **MBus decoders** and specialized **rule nodes (Loriot, SOAP)**. Optimized high-throughput performance through strategic database management—leveraging **PostgreSQL**, **TimescaleDB**, and **Cassandra**—while utilizing **Kafka** and **RabbitMQ** to handle **millions of data** points with real-time **WebSocket** updates. Streamlined enterprise-grade delivery by orchestrating **Swisscom** integration, implementing a versatile export system, and deploying automated **CI/CD pipelines** via **Azure DevOps** to ensure high availability and scalability across the entire ecosystem.

Airtime Management System-AMS

2021 – 2021

Developed a secure internal web application for bKash Limited  by engineering core **REST APIs** featuring a robust **maker-checker approval workflow** and automated **email notification** systems. Optimized system performance and reliability by implementing **asynchronous schedulers**, centralized **Logback** configurations, and **Feign clients** for efficient inter-service communication. Built using **Java (Spring Boot)** and **Oracle**, this solution streamlined administrative operations and ensured high data integrity through structured, workflow-driven backend processes.

MULTIBANK

2020 – 2021

Built an internal web application for **bKash Limited** to streamline and automate key backend processes. The system supports **SFTP inbound** and **outbound** file transfers for secure data exchange. It includes configurable **email notifications** implemented using **asynchronous** processing for improved performance. A **maker-checker approval** workflow was implemented to ensure controlled and validated operations. The solution was built using **Java Spring Boot**, **Spring Integration**, **APIs**, **jQuery**, and **MySQL** with **logback** integration for centralized logging.

PREVENTO

2019 – 2020

Developed a comprehensive personalized **healthcare** platform using **PHP (Laravel)** and **MySQL** that generates customized monthly care plans based on user health metrics, including **biometrics**, **diagnosed illnesses**, and **risk factors**. Architected a secure communication bridge enabling patients to **message** medical professionals directly, while integrating **Firebase** for **real-time engagement** and **AWS** for scalable hosting. Enhanced platform insights and **user tracking** through strategic implementation of **Google Analytics**, ensuring a **data-driven** approach to delivering actionable health tasks and professional medical consultation services.

EDUCATION

Bachelor of Science in CSE

International Islamic University Chittagong 

03/2014 – 11/2019
Chittagong,
Bangladesh

OPEN-SOURCE CONTRIBUTION

ThingsBoard

<https://github.com/thingsboard/thingsboard>

PR: 6614 